Faculty of Sciences

Department of Computer Sciences

Master thesis

Procedural Generation of Solvable Levels in Multi-Agent Reinforcement Learning Environment

Author:
Charels Hugo
hugo.charels@ulb.be

Supervisors:
Lenaerts Tom
Molinghen Yannick

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Contents

1 Introduction 1
1.1 Observations 1
2 Background 2
3 Related Work 3
4 Methods 4
4.1 Solver by Reduction to SAT 4
4.1.1 Definition of Sets 4
4.1.2 Definition of Variables 4
4.1.3 Constraints and Logical Encoding 4
5 Experiments 8
6 Conclusion 9
Bibliography 10

Chapter 1

Introduction

1.1 Observations

Chapter 2

Background

Chapter 3

Related Work

Chapter 4

Methods

4.1 Solver by Reduction to SAT

4.1.1 Definition of Sets

We consider a rectangular grid of height H and width W. Positions are identified by pairs (x, y), where $x \in \{0, ..., W-1\}$ denotes the column index and $y \in \{0, ..., H-1\}$ denotes the row index. The following sets are used throughout.

- $C = \{0, 1, ..., n_a - 1\}$: set of agent colors, where $n_a \geq 1$ is the number of agents.
- $D = \{N, S, E, W\}$: set of laser beam directions (North, South, East, West).
- $P = \{(x, y) \mid 0 \leq x < W, 0 \leq y < H\}$: set of all grid positions.
- $\mathcal{W} \subseteq P$: set of wall positions.
- $\mathcal{S} \subseteq C \times D \times P$: set of laser sources; $(c, d, p) \in \mathcal{S}$ denotes a laser of color c shooting in direction d from position $p \in P$.
- $\mathcal{E} \subseteq P$: set of exit positions, with $|\mathcal{E}| = n_a$.
- $T = \{0, 1, ..., T_{\max}\}$: set of discrete time steps.

We also write $\mathrm{start}(c) \in P$ for the initial position of agent $c \in C$.

4.1.2 Definition of Variables

We introduce three families of propositional variables.

- $a_{c,x,y,t}$: true iff agent $c \in C$ occupies position $(x, y) \in P$ at time step $t \in T$.
- $b_{c,d,x,y,t}$: true iff the beam of laser source $(c, d, -) \in \mathcal{S}$ is active at position $(x, y) \in P$ at time $t \in T$.
- $l_{c,x,y,t}$: true iff a laser of color $c \in C$ is active at position $(x, y) \in P$ at time $t \in T$.

4.1.3 Constraints and Logical Encoding

We now formalize the constraints ensuring the correctness of the reduction. Each group of clauses corresponds to one logical component of the problem.

1. **Initialization**

   - **Agents:**
     Each agent $c \in C$ is placed at its designated starting position $\text{start}(c)$ at $t = 0$; all other positions are unoccupied by that agent:

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in P} \begin{cases} a_{c,x,y,0} & \text{if } (x,y) = \text{start}(c) \\ \neg a_{c,x,y,0} & \text{otherwise} \end{cases}$$

   - **Laser sources:**
     For each laser source $(c, d, (x_s, y_s)) \in \mathcal{S}$, the beam is always active at its origin, at every time step:

$$\bigwedge_{(c,d,(x_s,y_s)) \in \mathcal{S}} \bigwedge_{t \in T} b_{c,d,x_s,y_s,t}$$

2. **Agent Movements**

   We define the set of positions reachable from (x, y) in one step as (x, y) itself together with its four grid neighbors, excluding walls and laser source positions:

$$\begin{aligned} \text{next}(x,y) = \{(x',y') \in \{(x,y),(x,y-1),(x+1,y),(x,y+1),(x-1,y)\} \mid \\ (x',y') \in P, (x',y') \notin \mathcal{W}, (x',y') \notin \{p \mid (c,d,p) \in \mathcal{S}\}\} \end{aligned}$$

   - **Forward consistency:**
     If agent c is at position (x, y) at time t, it must be at some position in $\text{next}(x, y)$ at time $t + 1$:

$$\bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} a_{c,x,y,t} \rightarrow \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t+1}$$

$$\Updownarrow$$

$$\bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} \neg a_{c,x,y,t} \vee \bigvee_{(x',y') \in \text{next}(x,y)} a_{c,x',y',t+1}$$

   Two methods are proposed to enforce that each agent occupies at most one position at each time step.

   - **Global uniqueness:**
     No two distinct positions can both be occupied by agent c at time t:

$$\bigwedge_{c \in C} \bigwedge_{t \in T} \bigwedge_{(x_1,y_1) \in P} \bigwedge_{\substack{(x_2,y_2) \in P, \\ (x_2,y_2) \neq (x_1,y_1)}} \neg a_{c,x_1,y_1,t} \vee \neg a_{c,x_2,y_2,t}$$

   - **Local uniqueness:**
     No two distinct positions in $\text{next}(x, y)$ can simultaneously be occupied by agent c at time $t + 1$:

$$\bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} \bigwedge_{(x',y') \in \text{next}(x,y)} \bigwedge_{\substack{(x'',y'') \in \text{next}(x,y), \\ (x'',y'') \neq (x',y')}} \neg a_{c,x',y',t+1} \vee \neg a_{c,x'',y'',t+1}$$

Backward consistency:
If agent c is at position (x, y) at time $t+1$, it must have been at some position in $\text{next}(x, y)$ at time t:

$$\bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} a_{c,x,y,t+1} \rightarrow \bigvee_{(x',y') \in \text{ next}(x,y)} a_{c,x',y',t}$$

$$\Updownarrow$$

$$\bigwedge_{c \in C} \bigwedge_{t=0}^{T_{\max}-1} \bigwedge_{(x,y) \in P} \neg a_{c,x,y,t+1} \vee \bigvee_{(x',y') \in \text{ next}(x,y)} a_{c,x',y',t}$$

- **No simultaneous occupation:**
  Two distinct agents cannot share the same position at the same time, nor can they move to a position already occupied by the other agent:

$$\bigwedge_{c_1 \in C} \bigwedge_{c_2 \in C, c_2 \neq c_1} \bigwedge_{(x,y) \in P} \bigwedge_{t \in T} \neg a_{c_1,x,y,t} \vee \neg a_{c_2,x,y,t}$$

$$\bigwedge_{c_1 \in C} \bigwedge_{c_2 \in C, c_2 \neq c_1} \bigwedge_{(x,y) \in P} \bigwedge_{t=0}^{T_{\max}-1} \neg a_{c_1,x,y,t+1} \vee \neg a_{c_2,x,y,t}$$

$$\bigwedge_{c_1 \in C} \bigwedge_{c_2 \in C, c_2 \neq c_1} \bigwedge_{(x,y) \in P} \bigwedge_{t=0}^{T_{\max}-1} \neg a_{c_1,x,y,t} \vee \neg a_{c_2,x,y,t+1}$$

- **Victory condition:**
  Each exit must be occupied by at least one agent at time $T_{\max}$. Since $|\mathcal{E}| = n_a$ and agents cannot share positions, this enforces a bijection between agents and exits:

$$\bigwedge_{(x,y) \in \mathcal{E}} \bigvee_{c \in C} a_{c,x,y,T_{\max}}$$

- **Stay on exit:**
  Once an agent reaches an exit, it remains there for all subsequent time steps:

$$\bigwedge_{c \in C} \bigwedge_{(x,y) \in \mathcal{E}} \bigwedge_{t=0}^{T_{\max}-1} \neg a_{c,x,y,t} \vee a_{c,x,y,t+1}$$

3. **Laser Activity**

We define $\text{next}_{d(x,y)}$ as the position immediately adjacent to (x, y) in direction d:

$$\text{next}_d(x, y) = \begin{cases} (x, y-1) \text{ if } d = N \\ (x+1, y) \text{ if } d = E \\ (x, y+1) \text{ if } d = S \\ (x-1, y) \text{ if } d = W \end{cases}$$

- **Walls block beams:**
  A beam cannot be active at a wall position:

$$\bigwedge_{(c,d,-) \in \mathcal{S}} \bigwedge_{(x,y) \in \mathcal{W}} \bigwedge_{t \in T} \neg b_{c,d,x,y,t}$$

- **Beam propagation:**
  For each laser source $(c, d, -) \in \mathcal{S}$ and each non-wall position (x, y) whose successor $(x', y') = \text{next}_{d(x,y)}$ also lies outside $\mathcal{W}$, the beam propagates from (x, y) to (x', y') if and only if no agent of color c is present at (x', y'):

$$\bigwedge_{(c,d,-)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_d(x,y)\in P\setminus\mathcal{W}}} b_{c,d,x',y',t} \leftrightarrow \left(b_{c,d,x,y,t} \wedge \neg a_{c,x',y',t}\right)$$

$$\Updownarrow$$

$$\bigwedge_{(c,d,-)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_d(x,y)\in P\setminus\mathcal{W}}} \neg b_{c,d,x,y,t} \vee a_{c,x',y',t} \vee b_{c,d,x',y',t}$$

$$\bigwedge_{(c,d,-)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_d(x,y)\in P\setminus\mathcal{W}}} b_{c,d,x,y,t} \vee \neg b_{c,d,x',y',t}$$

$$\bigwedge_{(c,d,-)\in\mathcal{S}} \bigwedge_{t\in T} \bigwedge_{\substack{(x,y)\in P\setminus\mathcal{W},\\ \text{next}_d(x,y)\in P\setminus\mathcal{W}}} \neg a_{c,x',y',t} \vee \neg b_{c,d,x',y',t}$$

- **Link between beam and laser variables:**
  For each laser source $(c, d, -) \in \mathcal{S}$, the laser variable $l_{c,x,y,t}$ is true at a position if and only if the beam of that source is active there:

$$\bigwedge_{(c,d,-)\in\mathcal{S}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} b_{c,d,x,y,t} \leftrightarrow l_{c,x,y,t}$$

$$\Updownarrow$$

$$\bigwedge_{(c,d,-)\in\mathcal{S}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \left(\neg b_{c,d,x,y,t} \vee l_{c,x,y,t}\right) \wedge \left(\neg l_{c,x,y,t} \vee b_{c,d,x,y,t}\right)$$

- **Agents cannot step on active lasers:**
  An agent of color c_1 cannot occupy a position where a laser of color $c_2 \neq c_1$ is active. Agents are immune only to lasers of their own color:

$$\bigwedge_{c_1\in C} \bigwedge_{\substack{c_2\in C,\\ c_2\neq c_1}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} l_{c_2,x,y,t} \rightarrow \neg a_{c_1,x,y,t}$$

$$\Updownarrow$$

$$\bigwedge_{c_1\in C} \bigwedge_{\substack{c_2\in C,\\ c_2\neq c_1}} \bigwedge_{(x,y)\in P} \bigwedge_{t\in T} \neg l_{c_2,x,y,t} \vee \neg a_{c_1,x,y,t}$$

Chapter 5

Experiments

Conclusion

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